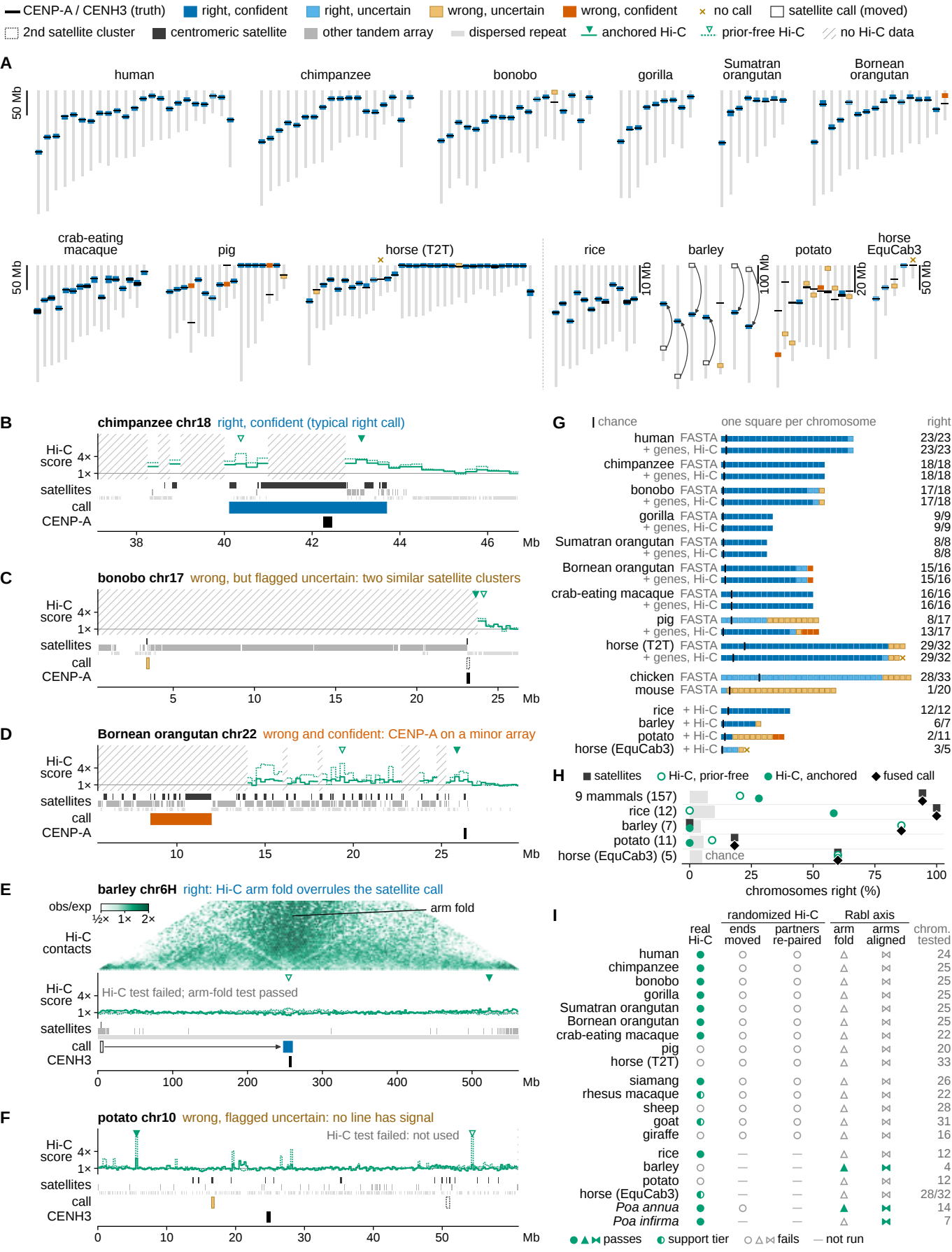

CENP-A / CENH3 (truth)
right, confident
right, uncertain
wrong, uncertain
wrong, confident
no call
satellite call (moved)
2nd satellite cluster
centromeric satellite
other tandem array
dispersed repeat
anchored Hi-C
prior-free Hi-C
no Hi-C data
A
human
chimpanzee
bonobo
gorilla
Sumatran orangutan
Bornean orangutan
crab-eating macaque
pig
horse (T2T)
rice
barley
potato
horse EquCab3
50 Mb
10 Mb
100 Mb
20 Mb
B
chimpanzee chr18 right, confident (typical right call)
C
bonobo chr17 wrong, but flagged uncertain: two similar satellite clusters
D
Bornean orangutan chr22 wrong and confident: CENP-A on a minor array
E
barley chr6H right: Hi-C arm fold overrules the satellite call
obs/exp
arm fold
Hi-C contacts
Hi-C test failed; arm-fold test passed
F
potato chr10 wrong, flagged uncertain: no line has signal
Hi-C test failed: not used
Hi-C score
satellites
call
CENP-A
CENH3
Mb
G
chance
one square per chromosome
right
human FASTA 23/23
+ genes, Hi-C 23/23
chimpanzee FASTA 18/18
+ genes, Hi-C 18/18
bonobo FASTA 17/18
+ genes, Hi-C 17/18
gorilla FASTA 9/9
+ genes, Hi-C 9/9
Sumatran orangutan FASTA 8/8
+ genes, Hi-C 8/8
Bornean orangutan FASTA 15/16
+ genes, Hi-C 15/16
crab-eating macaque FASTA 16/16
+ genes, Hi-C 16/16
pig FASTA 8/17
+ genes, Hi-C 13/17
horse (T2T) FASTA 29/32
+ genes, Hi-C 29/32
chicken FASTA 28/33
mouse FASTA 1/20
rice + Hi-C 12/12
barley + Hi-C 6/7
potato + Hi-C 2/11
horse (EquCab3) + Hi-C 3/5
H
satellites
Hi-C, prior-free
Hi-C, anchored
fused call
9 mammals (157)
rice (12)
barley (7)
potato (11)
horse (EquCab3) (5)
chance
chromosomes right (%)
I
randomized Hi-C
Rabl axis
real Hi-C
ends moved
partners re-paired
arm fold
arms aligned
chrom. tested
human 24
chimpanzee 25
bonobo 25
gorilla 25
Sumatran orangutan 25
Bornean orangutan 25
crab-eating macaque 22
pig 20
horse (T2T) 33
siamang 26
rhesus macaque 22
sheep 28
goat 31
giraffe 16
rice 12
barley 4
potato 12
horse (EquCab3) 28/32
Poa annua 14
Poa infirma 7
passes
support tier
fails
not run

Fusing satellite repeats with a Hi-C signal that must first prove itself places 148 of 157 mammal centromeres on CENP-A and rescues barley centromeres that satellites miss; randomized Hi-C never passes the test. Where evidence is thin, cenfuse says “uncertain” instead of guessing.

A Every chromosome with a mapped centromere, to scale: the call (colored box) against the measured CENP-A or CENH3 domain (black bar). In mammals and rice the two coincide; barley's calls move away from a misleading satellite; most potato centromeres are missed, and flagged.

B A typical right call: the call covers the centromeric satellite array that carries CENP-A.

C Two similar satellite clusters: cenfuse picks one, names the other as the alternative and says it is unsure; CENP-A sits on the alternative.

D The confident miss: CENP-A sits on a few small arrays far from the large array that was called.

E Barley's satellite marks distal chromosome regions, not centromeres. The arm fold of a Rabl nucleus, seen in Hi-C, moves the call onto CENH3.

F A potato centromere with nothing to follow: no satellite under CENH3 and no Hi-C signal; the call is wrong and flagged.

G Every chromosome as one square. Nearly all mammal centromeres are right from the DNA sequence alone, far above chance; adding genes and Hi-C mostly raises confidence (and restores pig). Chicken calls are right but uncertain; mouse calls land on the wrong satellite at the right end.

H Each evidence line alone. Satellites work in mammals and rice, Hi-C in barley, neither in potato; the fused call matches the best line in every genome.

I Hi-C acts only after it finds the other chromosomes' centromeres: in apes, crab-eating macaque, rice and both *Poa* species. Where it finds them only weakly (rhesus macaque, goat, horse EquCab3), it may only support calls. Randomized Hi-C passes no test. In Rabl nuclei (barley, *Poa*), folded and aligned arms test Hi-C instead.

Terms. Truth: CENP-A domains (centromere dip regions in great apes; CENP-A ChIP-seq in crab-eating macaque, pig and horse T2T, for pig and horse from other animals) and CENH3 or CENP-A ChIP-seq domains (rice, barley, potato, horse EquCab3). Right: the call overlaps the truth (mammal calls) or lies within four Hi-C bins of it (plant calls, and every line alone in H; bins 250 kb, human 500 kb, barley 2 Mb). Confident: confidence high or medium; uncertain: low. Chance: the probability that a call of the same length placed at random on the chromosome is right, summed over chromosomes. Runs: cenfuse 0.10.1, every benchmark arm rerun from the stored satellite searches. Every call is shown: “uncertain” is a designed output of the tool, and misses are shown on purpose.

A Gray bar: a chromosome hanging from its start; mammals and horse EquCab3 share one scale, rice, barley and potato have their own (scale bars). Box: final call with every input (mammals: FASTA, genes, Hi-C; plants: FASTA, Hi-C), colored by outcome, at least 0.75 mm tall; black bar: truth, at least 0.26 mm. Open boxes and arrows: barley's 6 satellite calls moved by Hi-C. Mammals 148/157 right (4 confidently wrong, 4 uncertain wrong, 1 no call; chance 7.1; $P = 6.0 \times 10^{-213}$, Poisson-binomial). Rice 12/12, barley 6/7, potato 2/11 (2 confidently wrong, 7 uncertain), horse EquCab3 3/5 (all uncertain); together 23/35, chance 2.3 ($P = 2.5 \times 10^{-20}$). Confident calls were right in 162 of 168 cases, uncertain or missing ones in 9 of 24 (Fisher's exact $P = 6.8 \times 10^{-12}$).

B–F Tracks: Hi-C score per bin, one scale for B–F (solid: anchored on the other chromosomes' satellite calls; dotted: prior-free; 1× = chromosome median; triangles: each line's peak; hatched: no Hi-C data, usually the unmappable array); satellite arrays by the role cenfuse gave them (dark: the family ranked centromeric; mid gray: other tandem arrays; light, thin: dispersed repeats, interspersed copies rejected as satellites; at least 0.12 mm wide); final call; truth. Windows: calls and truth ± 3 Mb (mammals), whole chromosome (plants). **B** was chosen by rule: of the great-ape right calls, the one nearest the medians of call width (3.3 Mb) and call-to-truth offset (0.45 Mb) of all 148 right mammal calls. **C** and **D** are the only great-ape misses. C: two small satellite clusters (22 and 23 kb); cenfuse took the first and reported the second, which lies under CENP-A and beside which both Hi-C peaks sit. D: CENP-A sits on 4 small arrays spanning 84 kb (73 kb of array) of families ranked centromeric, 14 Mb from the 3.4-Mb main array cluster that was called; only the anchored Hi-C peak points there, too weakly to overrule. **E**: the barley move of median length (lower median of 6); barley's satellite line follows a repeat family of distal chromosome regions, so the Hi-C test, anchored on those calls, fails; the arm-fold test passes and the call moves to the fold. Contact map: 836,986 cis read pairs, 4-Mb bins, balanced, observed/expected per distance, each pair of loci drawn above its midpoint, up to 300 Mb apart; in a Rabl nucleus the two arms lie side by side, so the fold appears as a ridge above the centromere. **F**: the median-distance miss among potato's 7 uncertain misses; small arrays of the centromeric family are scattered along the chromosome, none under CENH3, and Hi-C failed its test.

G One square per chromosome, sorted by outcome; black tick: chance expectation; numbers: right/total. Mammals, FASTA only: 143/157 right (13 of them uncertain), 13 wrong and uncertain, 1 confidently wrong; chance 10.5. Every input: 148/157 right (6 uncertain), 4 wrong and uncertain, 4 confidently wrong, 1 without a call; chance 7.1. Two of the uncertain right calls (bonobo chr14 and Bornean orangutan chr14) have a competing satellite cluster with no mappable DNA between it and the call, so Hi-C cannot tell the two apart. Pig from the FASTA alone: 8/17, all uncertain (no satellite family dominates, so every call is capped at uncertain); with genes 13/17, with genes and Hi-C 13/17 (3 of the 4 confident mammal misses; pig truth is weak ChIP-seq from other animals). Chicken (CENP-A windows on 33 of 40 chromosomes): 28/33, all uncertain. Mouse (the minor-satellite array; all chromosomes acrocentric): 1/20, all uncertain; every call sits at the centromeric end on the larger pericentric satellite, within 50 kb of the minor satellite on 20/20, which is the reported alternative on 20/20.

H Share of chromosomes right for each line alone (satellite call, prior-free Hi-C peak, anchored Hi-C peak) and for the fused call, every input, each within four Hi-C bins of the truth; gray band: chance for the most generous line. Mammals: satellites 148, anchored Hi-C 44, prior-free Hi-C 32, fused 148 of 157. Rice: satellites 12/12. Barley: prior-free Hi-C 6/7, satellites 0/7. Potato: no line above 2/11. Plants pooled: satellites 17/35, prior-free Hi-C 10/35, anchored Hi-C 10/35, fused 23/35.

I Hi-C test, per chromosome: the other chromosomes' Hi-C peaks must agree with their own confident calls on a majority, beyond chance (Poisson-binomial $P \le 0.01$); only then may Hi-C confirm, move or place calls. It passed on every tested chromosome of a genome or on none: in 8 of 14 mammals and in rice, *Poa annua* and *Poa infirma*. Half-filled: the support tier (agreement beyond chance but not on most chromosomes; Hi-C may then only raise a call its own peak agrees with), in rhesus macaque, goat and horse EquCab3 (on 28 of 32 chromosomes); it supported 5 rhesus macaque calls already high, raised 8 goat calls to high and supported none in horse EquCab3. Elsewhere Hi-C only reports. Randomized Hi-C (read ends moved, or partners re-paired at random): 0 of 708 chromosome tests passed in 29 runs, none reached the support tier, no Rabl-axis test passed and no call changed; likewise in the 18 other randomized runs of the benchmark. Rabl axis (real Hi-C): the arm fold (each chromosome's own map folds at its centromere) passes in barley and *Poa annua*; the trans arm alignment (arms of different chromosomes align by distance from their centromeres) in barley, *Poa annua* and *Poa infirma* (not acting in *Poa infirma*, whose arm fold does not lean the same way, P 0.23). Numbers: chromosomes tested (28/32: in the support tier). Horse EquCab3 has no confident call, so its uncertain calls serve as references at clean peaks. Mammals without functional truth appear only here.

Exceptions. Horse T2T is 29/32 from the FASTA alone and 29/32 with every input, but not on the same chromosomes: genes and Hi-C fix 2 uncertain calls (CM128058.1, CM128066.1) and lose 2 (CM128050.1, uncertain, 2.7 Mb from CENP-A; CM128070.1, uncertain, 2.6 Mb from CENP-A); chr11 (CM128059.1, a satellite-free centromere) gets no call with every input (5.6 Mb off from the FASTA alone). Horse EquCab3 makes no confident call. Examples: B, E and F by the stated median rules; C and D because they are the only great-ape misses, so the typical failure cannot be chosen among more.

Analysis: asc-oulab:/data2/chris/centromere_benchmark/v2/v3/v4/v5/v6/v7/v8/v9/v10/v10.1/v10.2/figures/benchmark_overview · Script: /data2/chris/centromere_benchmark/v2/v3/v4/v5/v6/v7/v8/v9/v10/v10.1/v10.2/figures/benchmark_overview/benchmark_overview.py · 2026-11-04